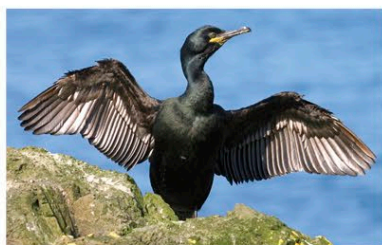


BIRDS

Birds have busy, active lives. Many are spectacularly beautiful; others sing complex – even musical – songs. They have the intelligence and parental devotion of a mammal combined with characteristics more reminiscent of a reptilian ancestor.

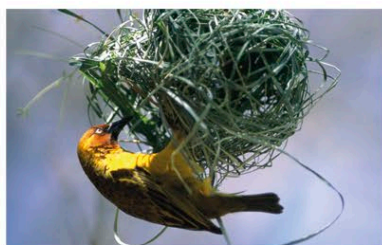
PHYLUM	CHORDATA
CLASS	AVES
ORDERS	40
FAMILIES	250
SPECIES	10,770



Asymmetrical flight feathers, with a narrow outer blade and broader inner blade, give birds better control in the air.



Many chicks are altricial – born in a helpless state. Blind and naked, they require prolonged parental care.



Elaborate nests are built by the male weaver bird to impress prospective mates, testament to the brain power of birds.

DEBATE

ARE BIRDS DINOSAURS?

Traditionally birds are in a class separate from reptiles, including dinosaurs. But today scientists hold that all descendants of a common ancestor should be classified together to reflect evolutionary relationships – a method known as cladistics. This makes birds as much a part of the dinosaur group as *Tyrannosaurus*.

Birds are the only living animals with feathers. They are warm-blooded vertebrates that stand on two legs and their forelimbs are considerably modified as wings. As well as facilitating flight, feathers insulate the body, so many birds stay active in icy-cold conditions, just like furry mammals. Feathers can also be colourful, showing that birds rely on flaunting visual cues to communicate and find a mate. Birds evolved from a group of two-legged, carnivorous dinosaurs that included *Tyrannosaurus*. It is possible – even likely – that these ancestors were already feathered.

TAKING TO THE AIR

No-one knows for sure how – or why – the first birds took to the air, but the fact that they did had effects that would be forever imprinted upon the bird's body. Their wristbones became fused together as arms evolved into wings. The breastbone evolved a huge keel for the attachment of larger muscles for powering the flapping wings. Already their dinosaur ancestors had air-filled, but strong bones. But birds coupled this with a strong cardiovascular system powered by a mammal-like four-chambered heart and a rapid metabolism to generate lots of energy. A system of air sacs – linked to the bone chambers – expanded their breathing capacity to flush stale air out of lungs more efficiently than in mammals.

Many reptilian features still remain. The bare parts of a bird's legs and toes have horny reptilian scales. Birds excrete a semi-solid waste product called uric acid (not a solution of urea, as in mammals). Waste from the kidneys and gut mix and exit via a common reptilian opening called a cloaca. But a bird's brain is better developed than that of a reptile, because it is supported by a warm-blooded body and fast metabolism. This makes birds not only skilled at flying, but remarkably sophisticated in the ways they acquire food or raise a family. Young birds hatch from reptilian hard-shelled eggs, but they are reared by intelligent parents who will invest much time and energy into raising them to maturity. Sixty million years of evolution mean that a bird is no longer just a reptile with feathers.

CROWNING GLORY >

Birds' remarkable colours and structures often appear over extravagant, such as this pigeon's elaborate head crest.

